

# Race Performance Report

SESSION

Race

LAYOUT

Daytona International Speedway Road Course

CAR CLASS

Hyper

TEAM

-

DURATION

23:06.720

DISTANCE

72.6 km

COMPOUNDS

0.0

SOC START / END

100.0 / 65.3

GENERATED

2026-07-30T23:16+02:00

CIRCUIT

Daytona International Speedway

CAR

AF Corse 2026 #83:LM

DRIVER

Driver unavailable

SESSION DATE

2026-07-30T21:47:51.747792

LAPS VALID / TOTAL

10 / 13

PIT ENTRIES / EXITS

1 / 1

FUEL START / END

88.0 / 50.8 L

FINISH / CLASSIFICATION

- / -

# Executive Summary / Sintesi

RECORDED / VALID LAPS

**13 / 10**

BEST VALID LAP

**1:35.354**

REPRESENTATIVE PACE

**1:35.734**

CONSISTENCY MAD

**0.308 s**

BEST SUSTAINED PACE

**1:35.734**

BEST 3 / 5 LAP AVG

**1:35.432 / 1:35.525**

FUEL / LAP

**2.955 L**

STINTS / PIT STOPS

**1 / 1**

CONFIDENCE

**High**

## Key findings

- Representative Pace: 1:35.734 (measured, high confidence). Median of 10 clean valid laps.
- Pace Trend: -0.110 s/lap (inferred, high confidence). Theil-Sen trend across clean laps.
- Fuel Effect: 0.037 s/L (inferred, medium confidence). robust pairwise estimate; fuel remains correlated with tyre age and track evolution
- Tyre Degradation: -0.110 s/lap (inferred, low confidence). inferred from within-stint clean-pace trend; fuel, traffic and track evolution cannot be fully separated

Position progression is shown only when measured; position changes are not called overtakes without opponent and pit-cycle evidence.

# Data Quality & Methodology

10 laps included and 3 excluded. Missing channels: position. Overall confidence: High. Excluded laps remain in the audit.

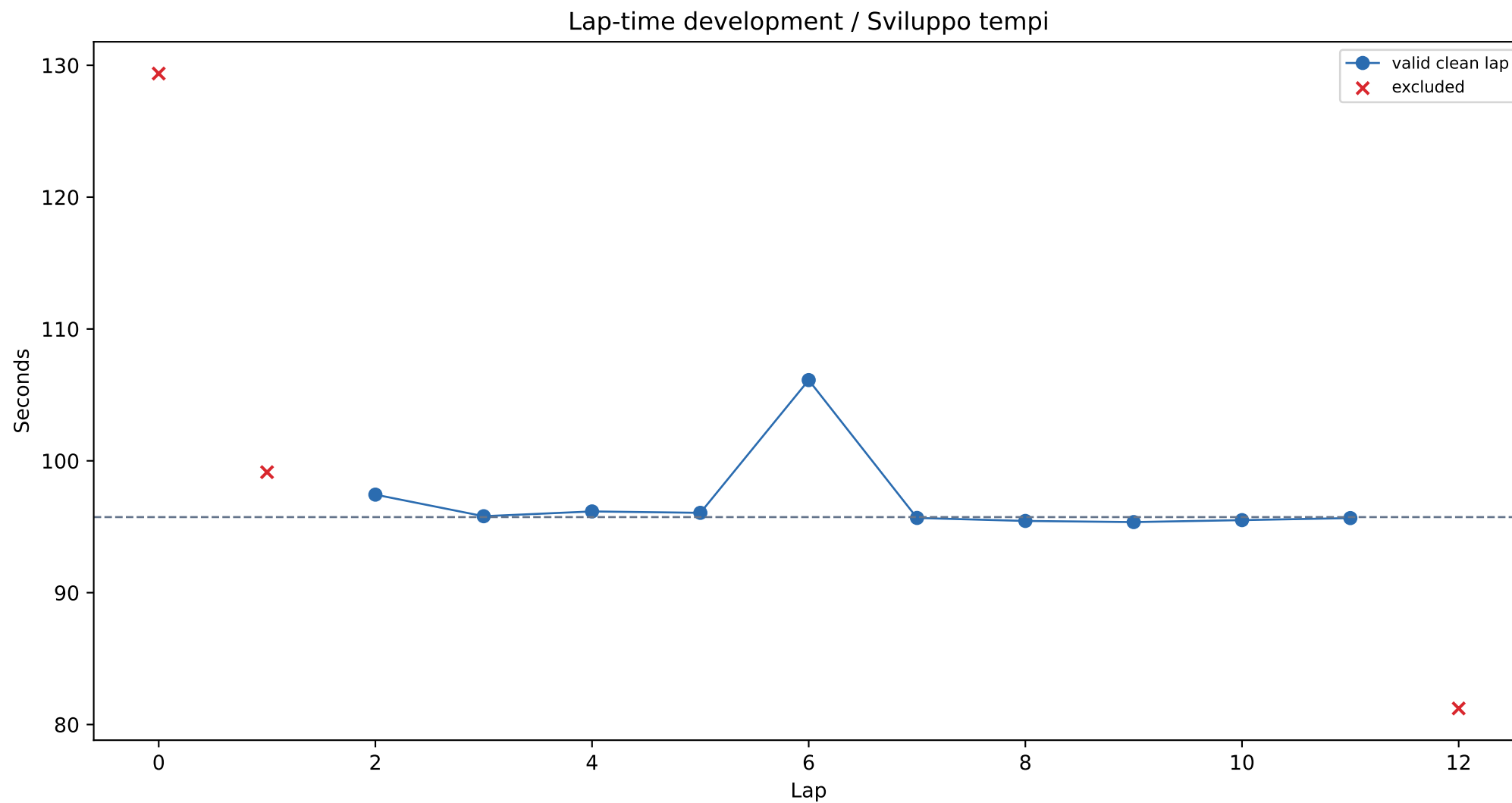
|                           |          |
|---------------------------|----------|
| Duplicate samples         | -        |
| Timestamp discontinuities | -        |
| Impossible samples        | -        |
| Samples analyzed          | 9906     |
| Missing channels          | position |
| Source warnings           | None     |

## Filters

shared lap-quality contract - finite timing 40-900 s - pit/yellow/incomplete exclusion - sensor anomaly audit

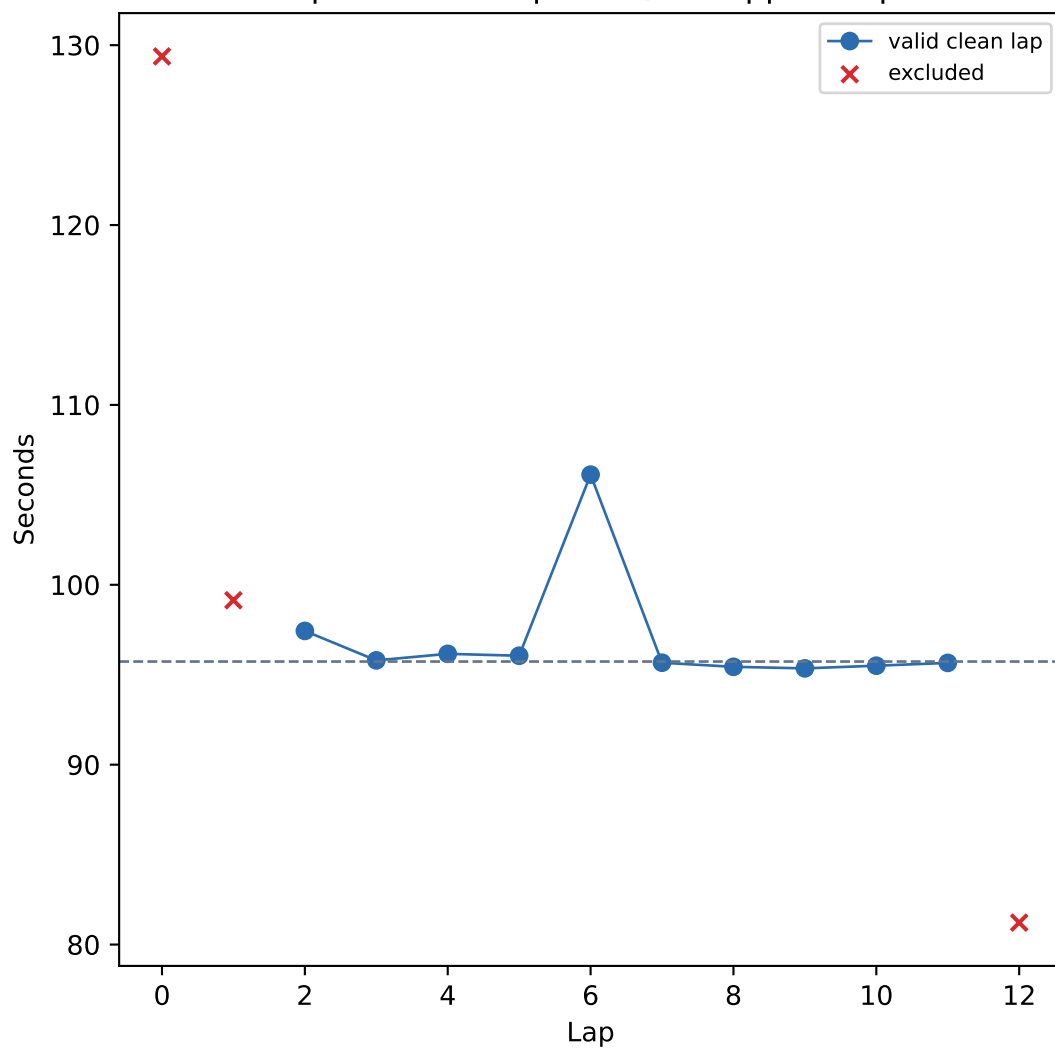
Methods: median, trimmed mean, MAD, Theil-Sen pairwise slope. Measured values and labeled inferences only; no external LLM.

# Session Timeline

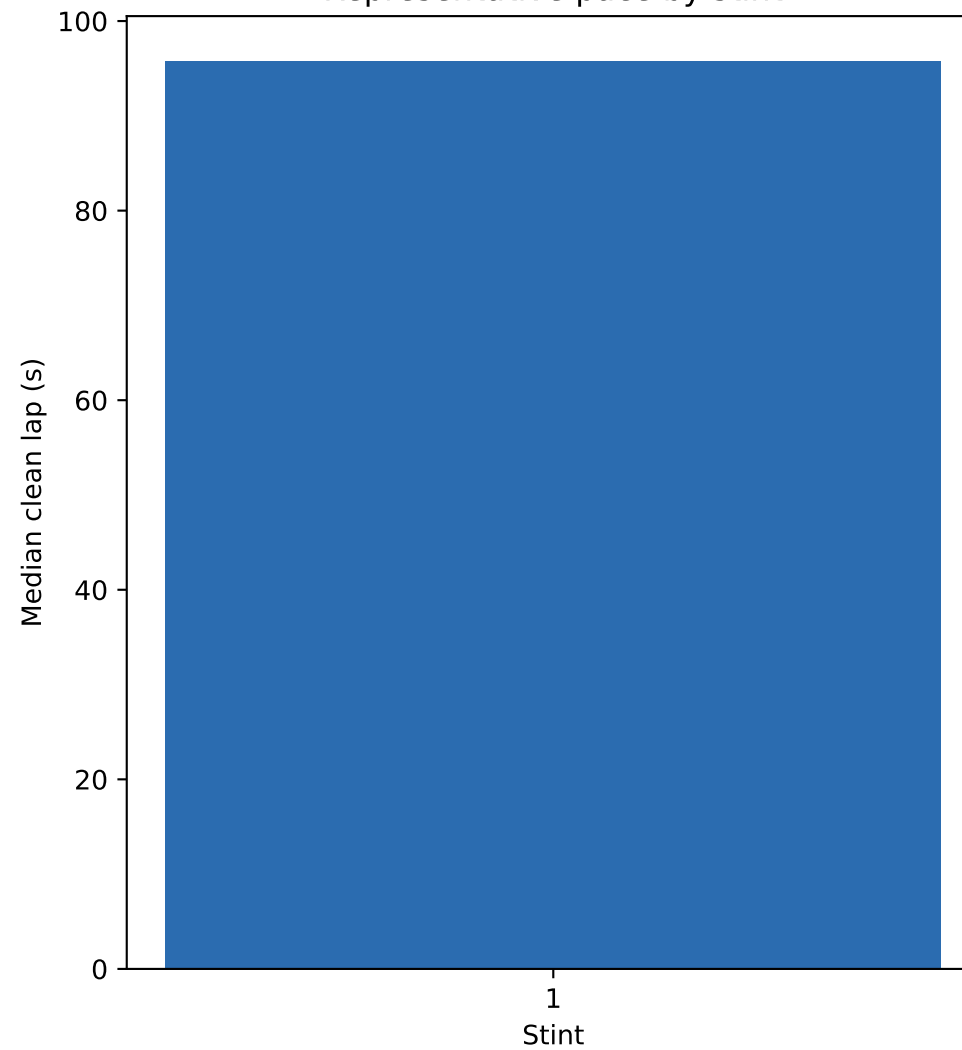


# Lap & Stint Analysis

## Lap-time development / Sviluppo tempi

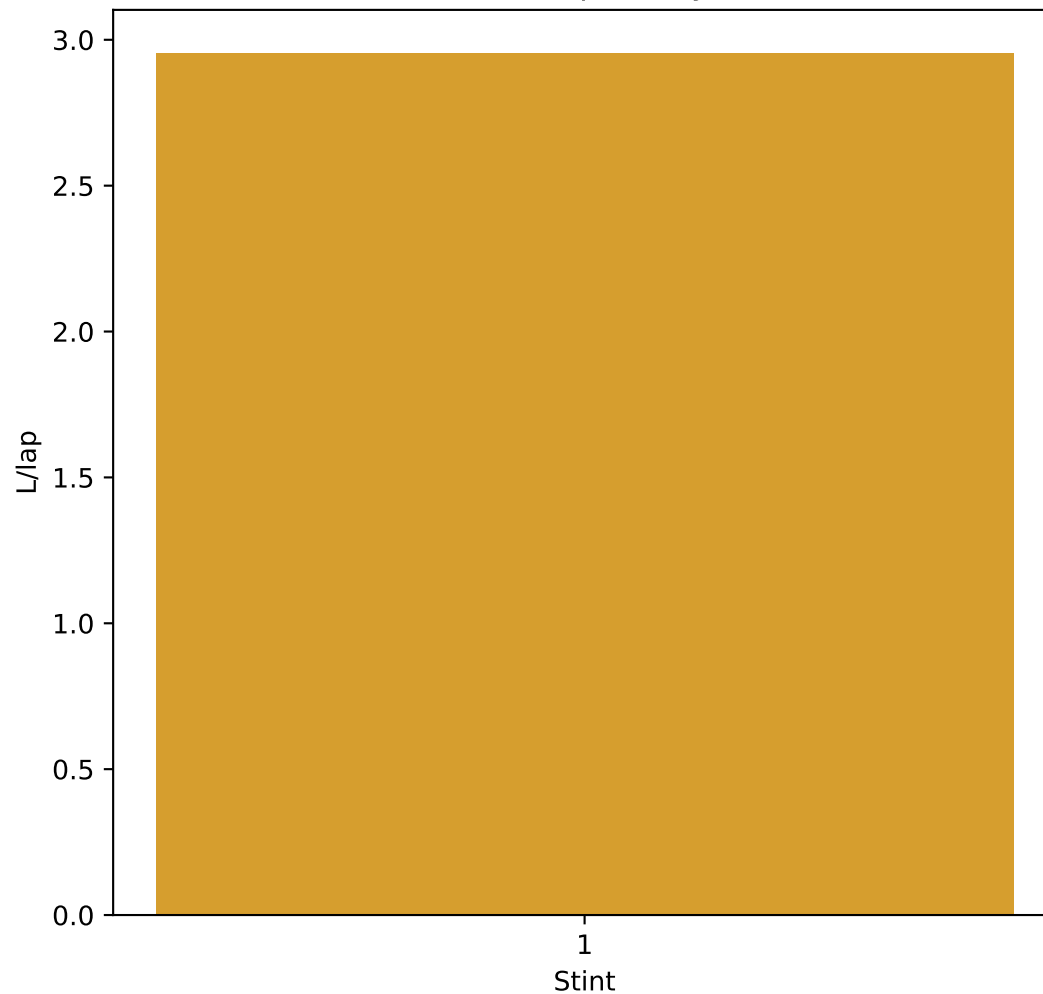


## Representative pace by stint



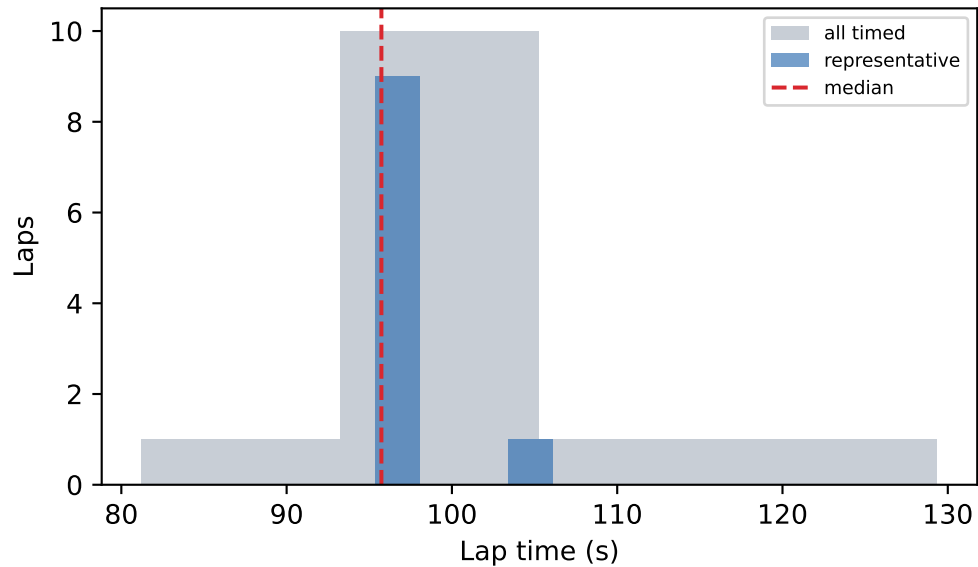
# Fuel & Tyre Analysis

Fuel consumption by stint



Fuel median: 2.955 L/lap Fuel effect: 0.0373 s/L (medium) robust pairwise estimate; fuel remains correlated with tyre age and track evolution Tyre degradation: -0.110 s/lap (low) inferred from within-stint clean-pace trend; fuel, traffic and track evolution cannot be fully separated

# Distribution, Energy, Thermal & Environment



## Robust distribution metrics

Mean (trimmed): 1:35.965    Median: 1:35.734    Std dev: 3.120 s    IQR: 0.594 s    MAD: 0.308 s

Best consecutive 3-lap average: 1:35.432    Best consecutive 5-lap average: 1:35.525

## Energy

SoC: 100.00 -> 65.26 Virtual energy:  
100.00 -> 56.52 Observed states only;  
deployment targets require vehicle-  
specific configuration.

## Thermal

Peak tyre: 81.2 C Peak brake: 575.3 C  
Oil max: 99.3 C; water max: 81.8 C  
Observed extremes; no ideal or  
warning threshold is claimed without  
vehicle-specific targets.

## Environment

Track temp change: 0.7 C Ambient  
change: 0.0 C Wetness change: 0.00

## Platform

Front ride height min: 0.028 Rear  
ride height min: 0.022 Absolute aero  
coefficients and bottoming thresholds  
are unavailable without vehicle  
definitions.

# Best Lap, Potential & Comparison

BEST ACTUAL

1:35.354

THEORETICAL BEST

-

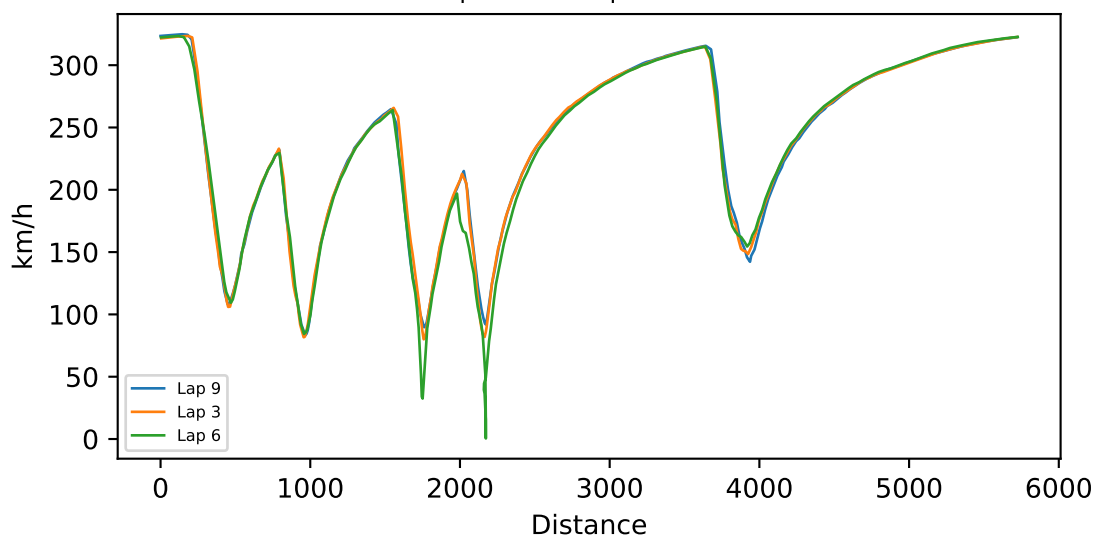
REALISTIC POTENTIAL

1:35.354

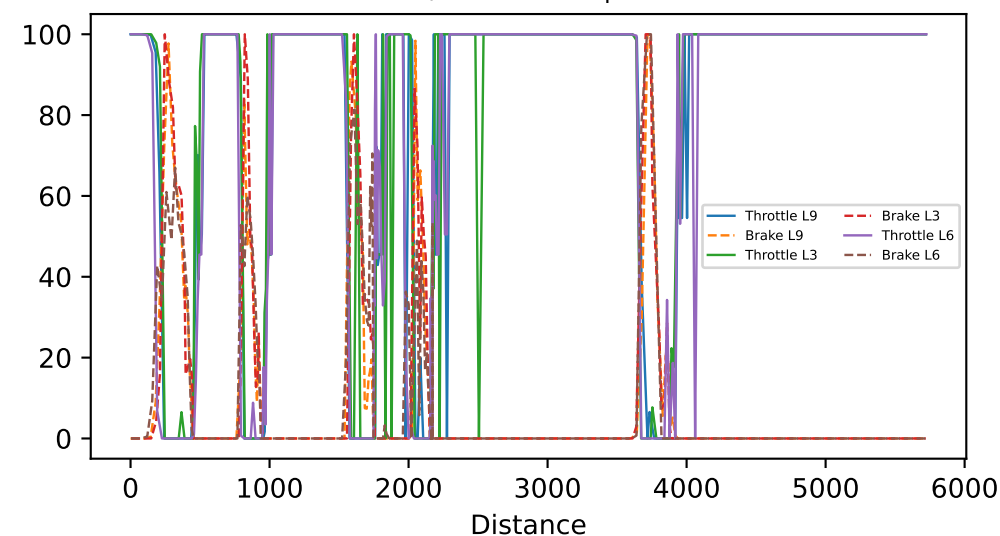
GAP

-

Speed over lap distance



Throttle / brake over lap distance



## Corner analysis

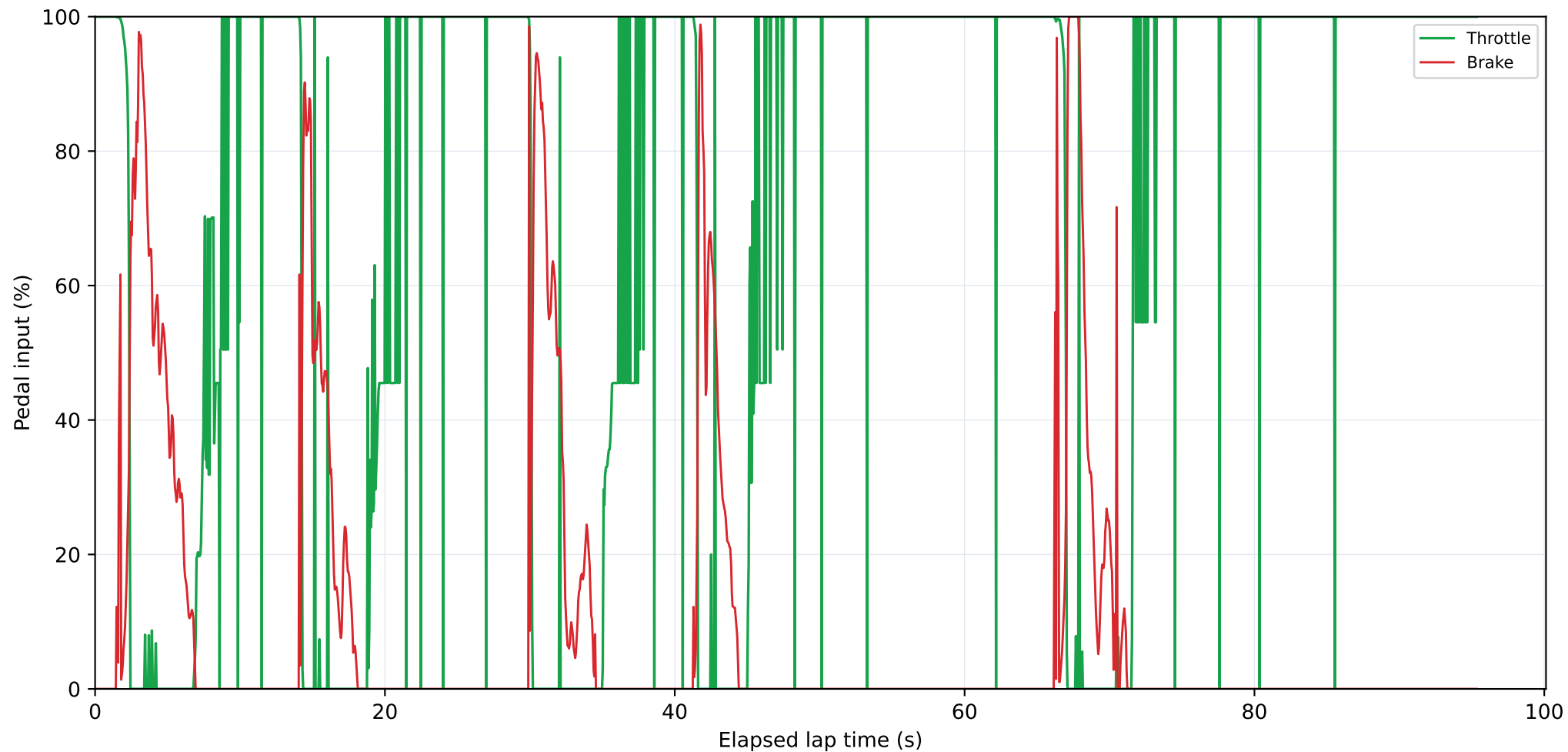
no shared corner map and compatible reference-lap segmentation are available

The theoretical best is not presented as automatically achievable. A realistic composite is only produced when compatible sector evidence exists.



# Best Lap Throttle & Brake - Lap 9

Best valid lap: 1:35.354. Inputs are aligned by elapsed lap time.



# Pits, Traffic & Event Evidence

## Pit stops

Detected: 1. opponent pit-cycle state is not recorded reliably

## Traffic

opponent gaps, identifiers and reliable position data are unavailable

## Position

position channel unavailable

## Corners

no shared corner map and compatible reference-lap segmentation are available

# Race Performance Assessment

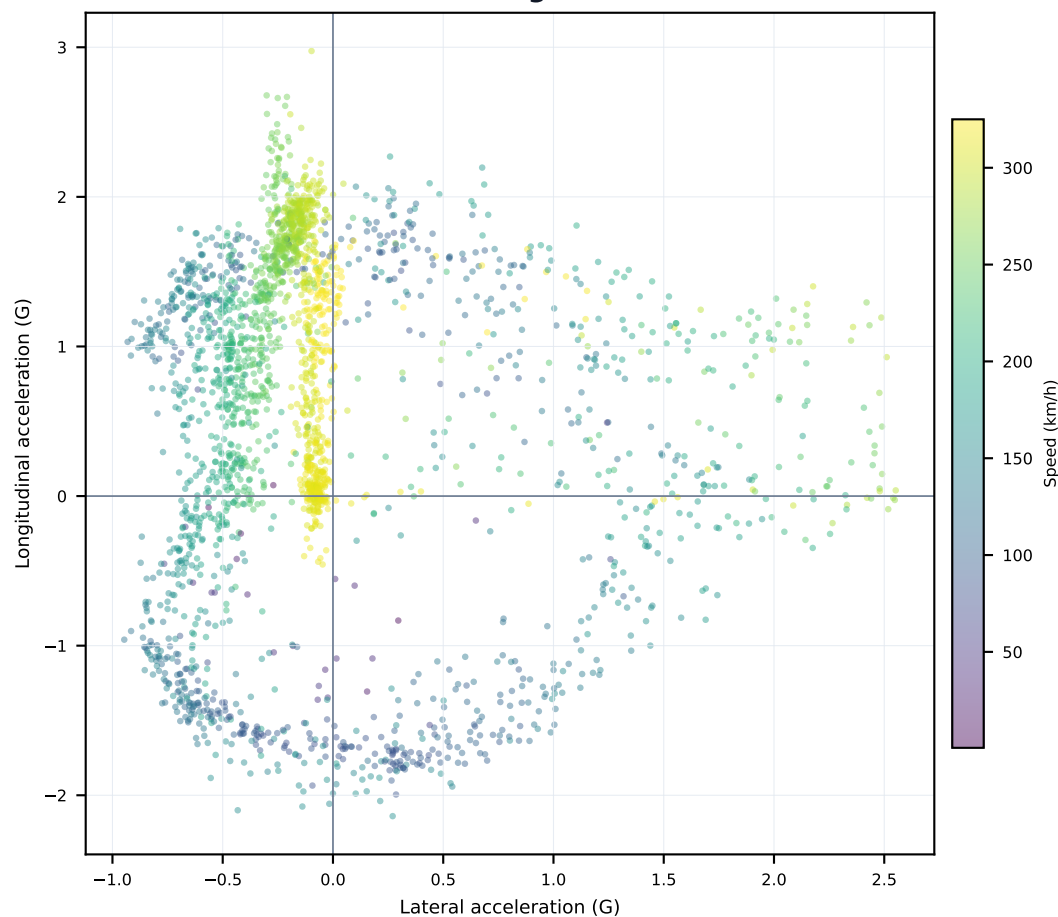
Position progression is shown only when measured; position changes are not called overtakes without opponent and pit-cycle evidence.

|                        |           |
|------------------------|-----------|
| Race Time              | 23:06.720 |
| Completed / Valid Laps | 13 / 10   |
| Distance               | 72.6 km   |
| Best Race Lap          | 1:35.354  |
| Median Clean Pace      | 1:35.734  |
| Best Stint             | 1         |
| Pit Stops              | 1         |
| Fuel Used              | 29.5 L    |

# Available X-Y Telemetry Plots

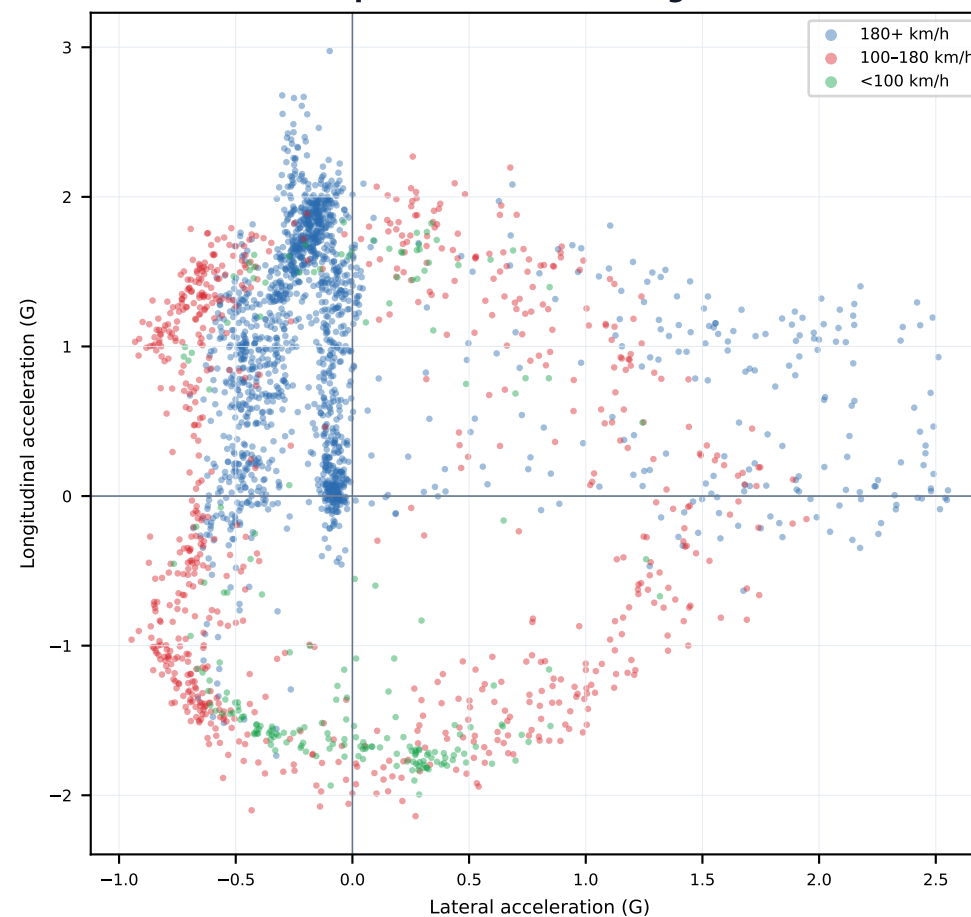
Plots use valid, non-pit laps and are included only when the required recorded channels produce data.

## G-G Diagram



7631 source points · Y min -2.239 · max 3.211 · avg 0.538 ·  $\sigma$  1.135

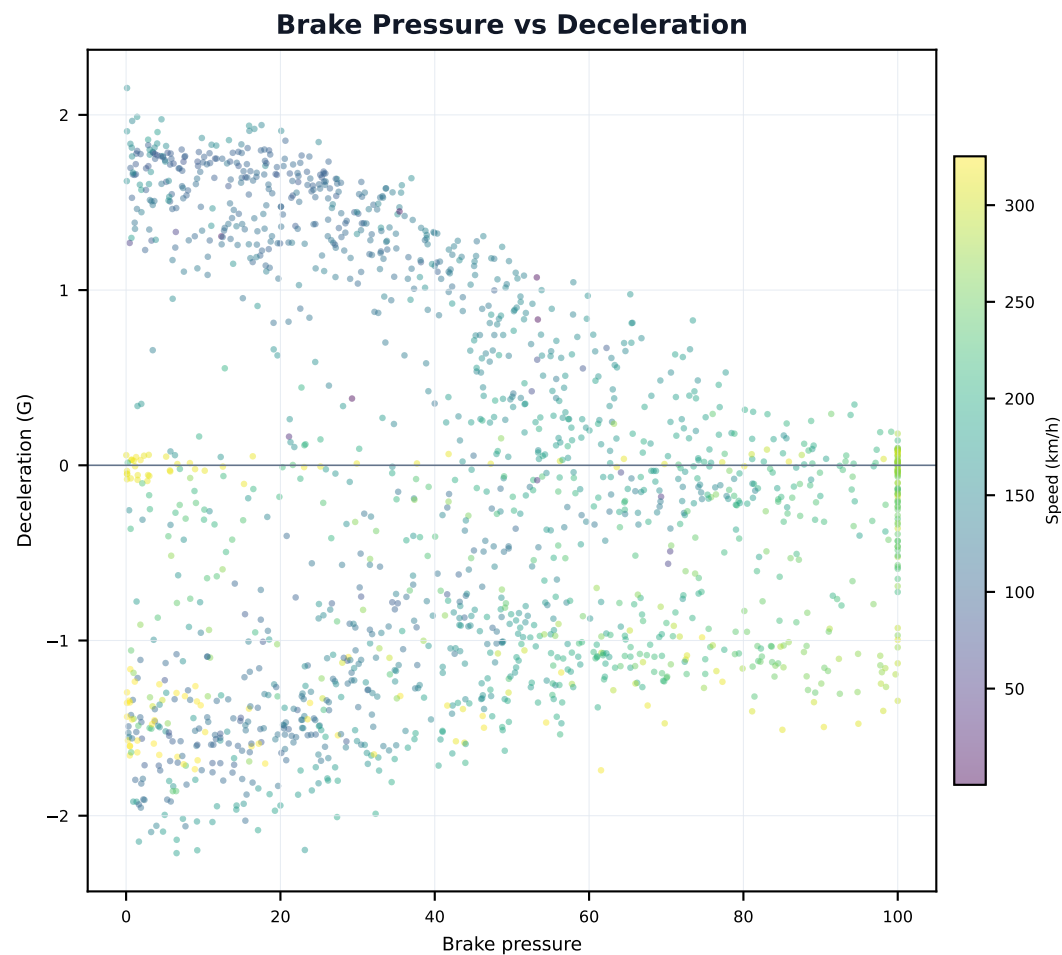
## Speed-Binned G-G Diagram



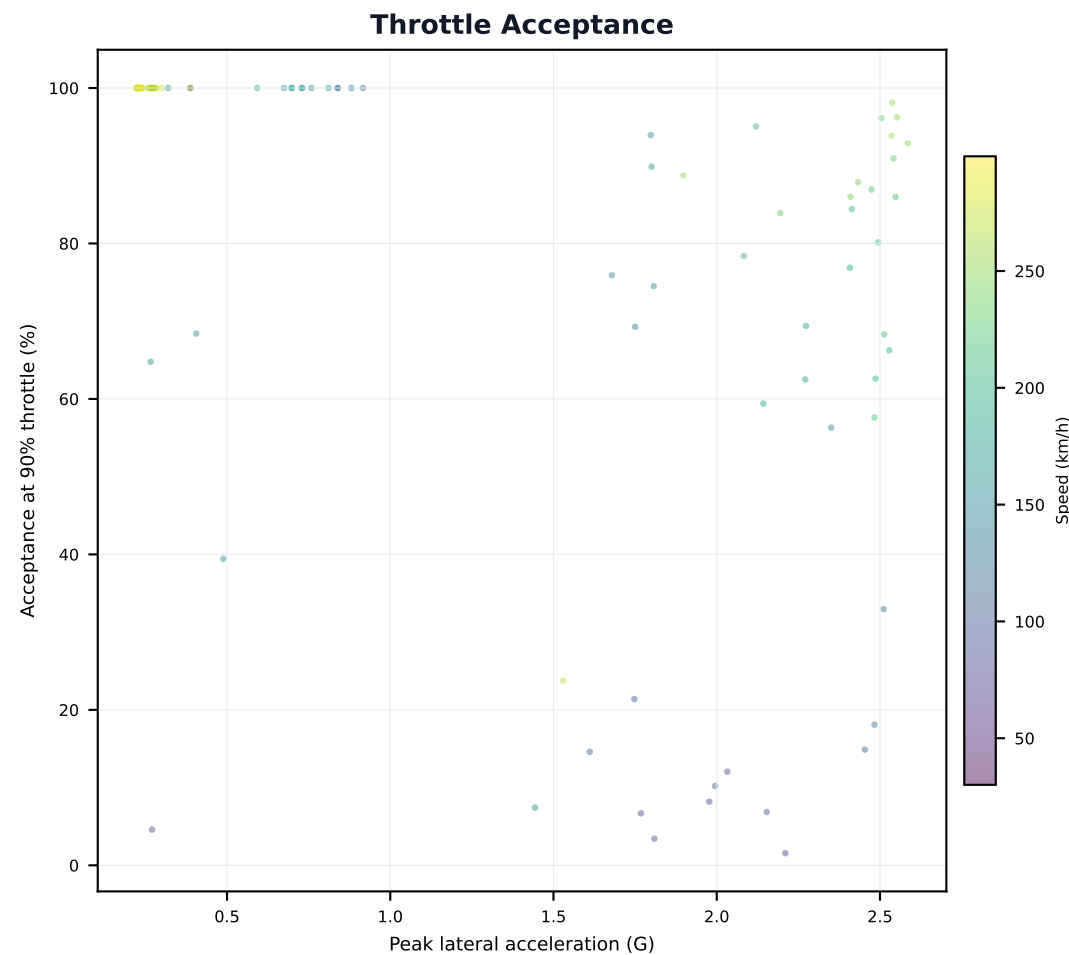
7631 source points · Y min -2.239 · max 3.211 · avg 0.538 ·  $\sigma$  1.135

# Available X-Y Telemetry Plots

Plots use valid, non-pit laps and are included only when the required recorded channels produce data.



1647 source points · Y min -2.214 · max 2.153 · avg -0.152 ·  $\sigma$  1.111

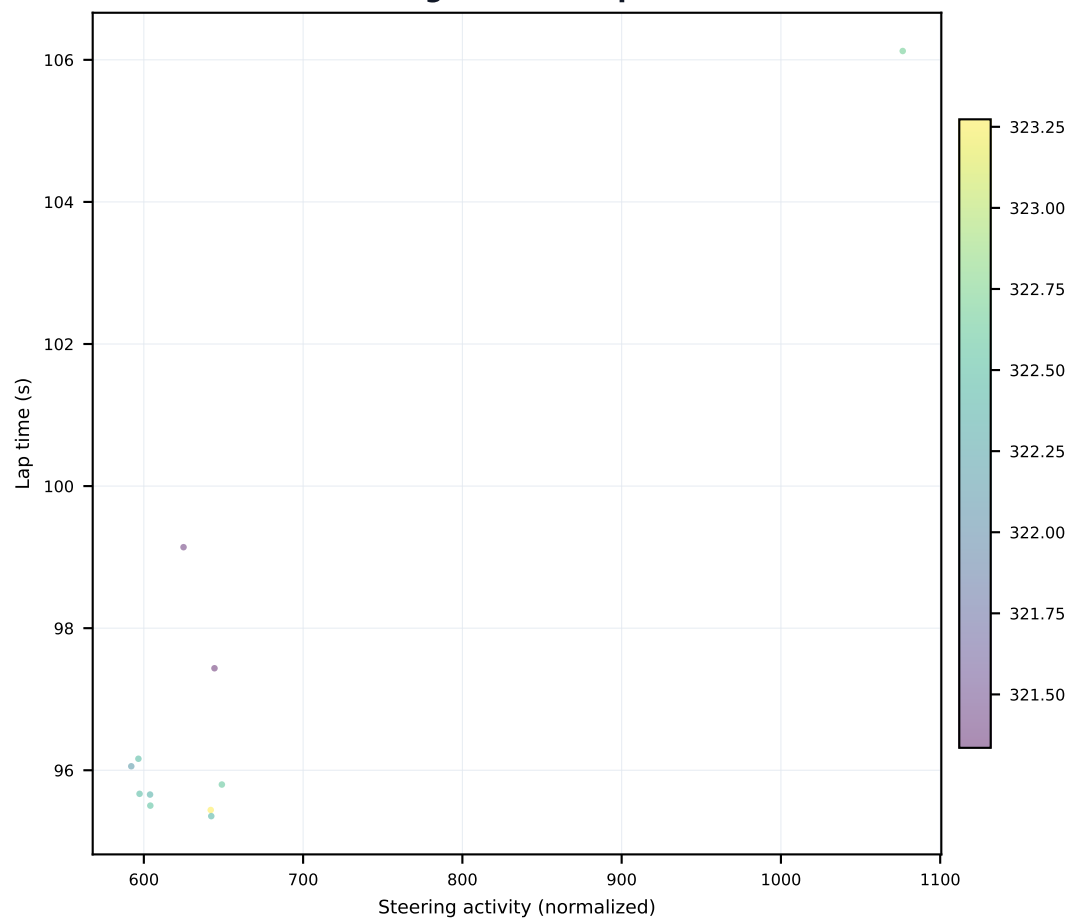


95 source points · Y min 1.566 · max 100.000 · avg 78.607 ·  $\sigma$  31.619

# Available X-Y Telemetry Plots

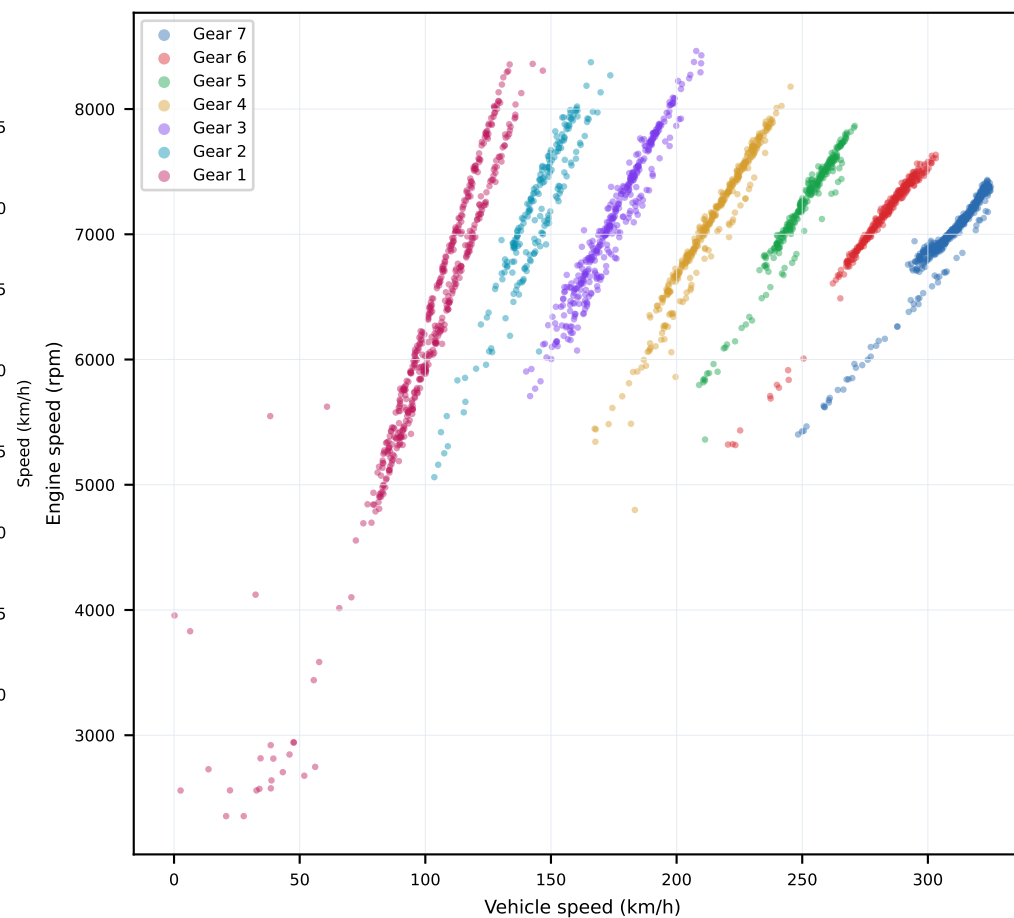
Plots use valid, non-pit laps and are included only when the required recorded channels produce data.

## Steering Work vs Lap Time



11 source points · Y min 95.354 · max 106.124 · avg 97.122 ·  $\sigma$  3.043

## Gear Chart

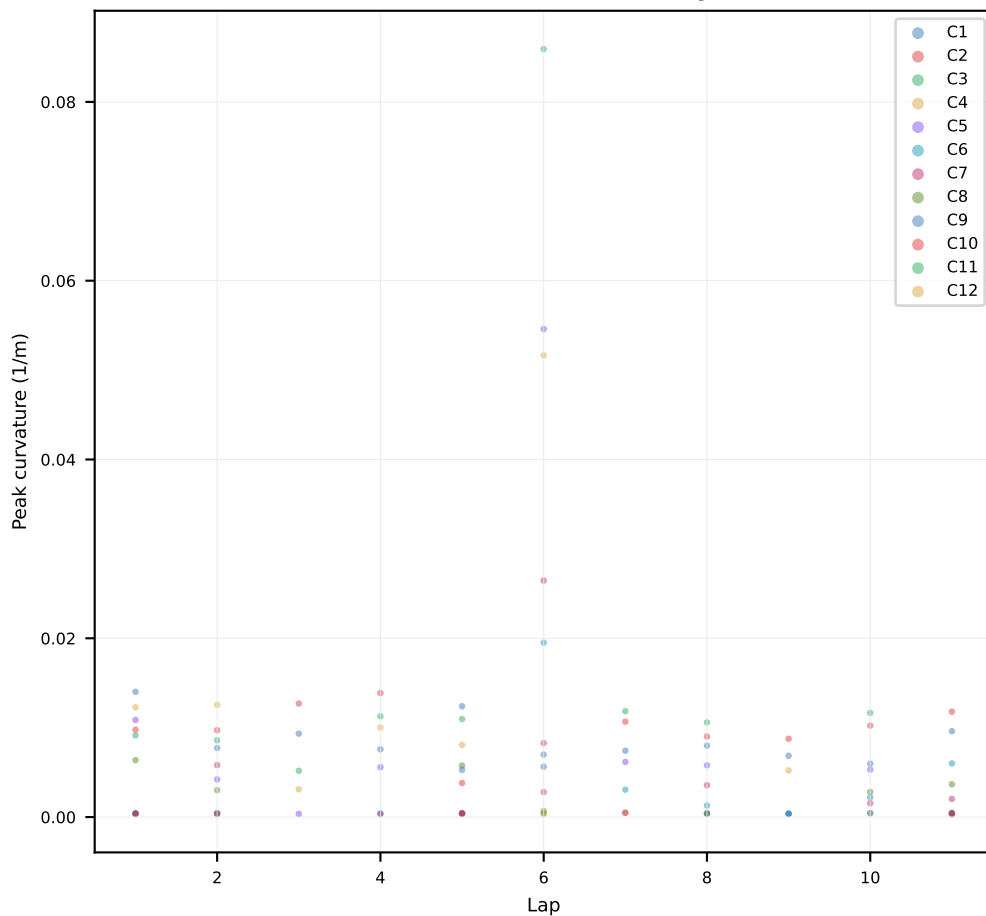


7505 source points · Y min 2353.543 · max 8494.205 · avg 6927.232 ·  $\sigma$  750.849

# Available X-Y Telemetry Plots

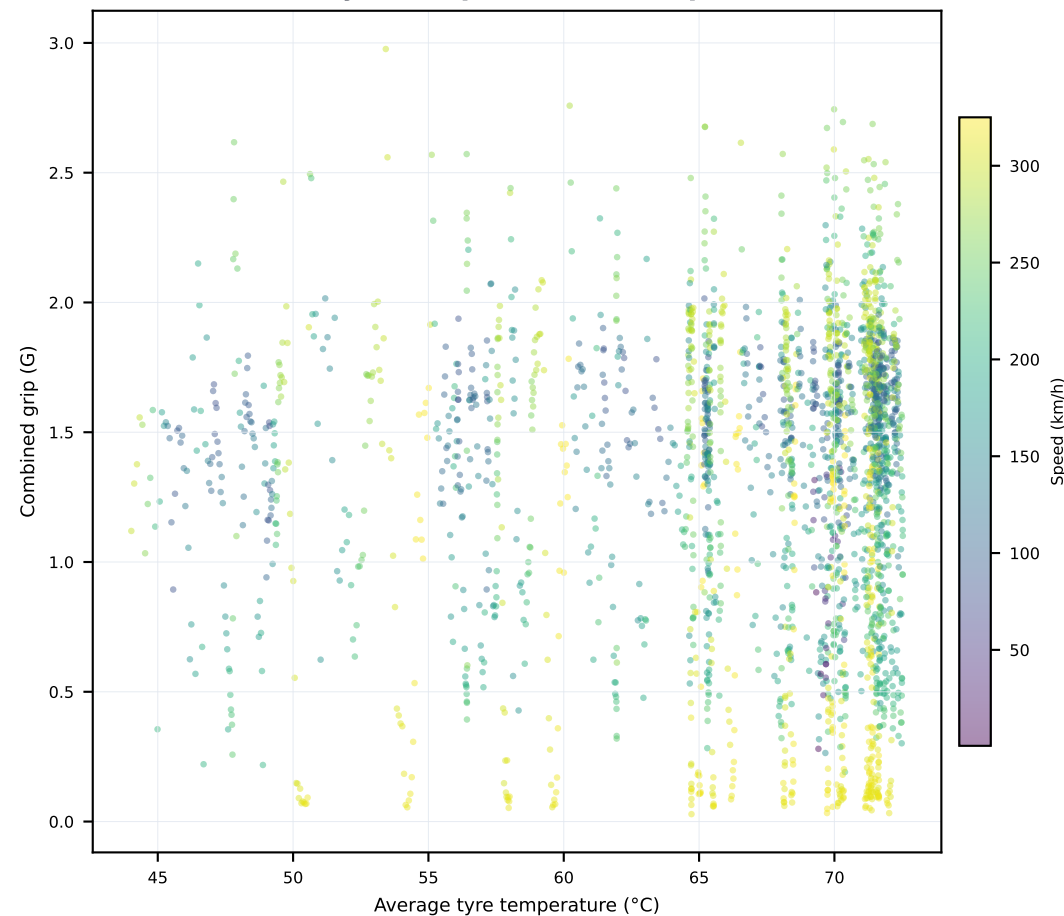
Plots use valid, non-pit laps and are included only when the required recorded channels produce data.

## Curvature Consistency



95 source points · Y min 0.000 · max 0.086 · avg 0.007 ·  $\sigma$  0.012

## Tyre Temperature vs Grip

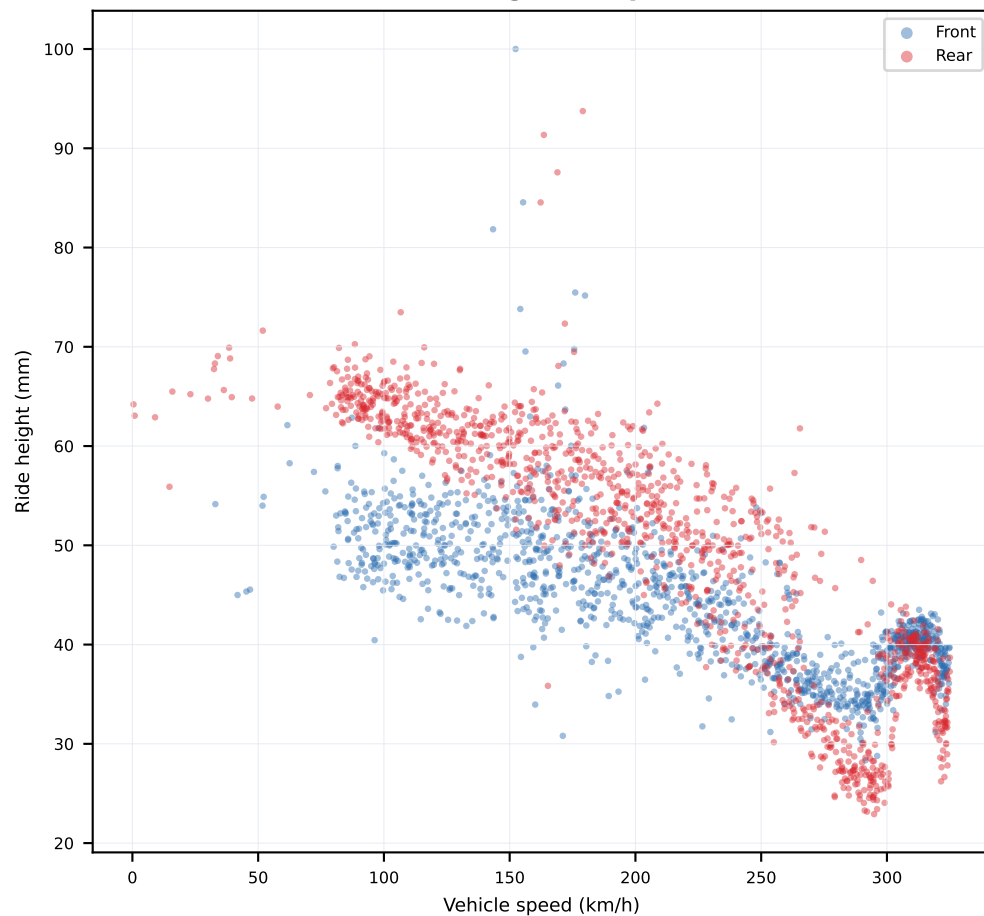


7631 source points · Y min 0.004 · max 3.213 · avg 1.309 ·  $\sigma$  0.578

# Available X-Y Telemetry Plots

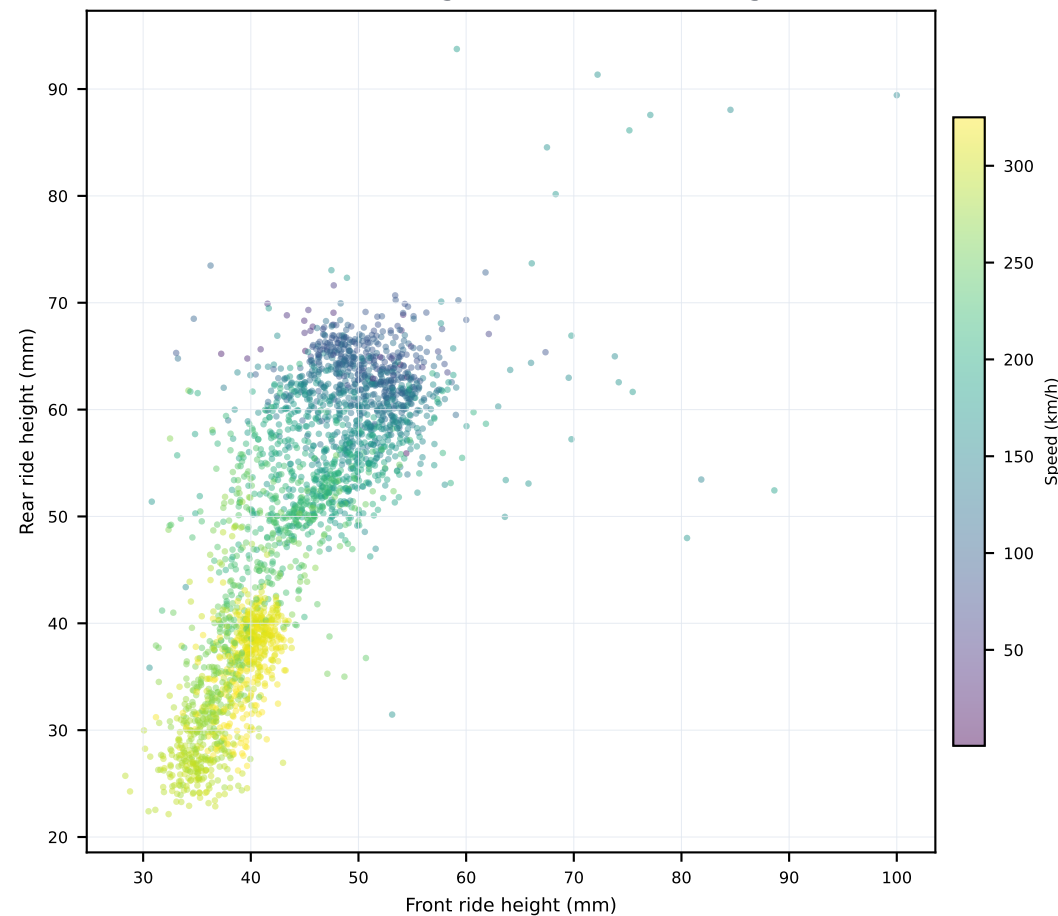
Plots use valid, non-pit laps and are included only when the required recorded channels produce data.

## Ride Height vs Speed



15262 source points · Y min 22.048 · max 100.000 · avg 46.750 ·  $\sigma$  10.634

## Front Ride Height vs Rear Ride Height



7631 source points · Y min 22.048 · max 95.065 · avg 49.112 ·  $\sigma$  12.866



# Complete Lap Table - laps 1 to 0

Every recorded lap is retained. A dash means the source channel or reliable derivation is unavailable.

| Lap | Time     | dBest  | dMed   | Classification               | St | Comp | Fuel S | Fuel E | Used | SoC S  | SoC E | Energy | Vmax | Tyre C | Brake C | Traffic | Off/Imp |
|-----|----------|--------|--------|------------------------------|----|------|--------|--------|------|--------|-------|--------|------|--------|---------|---------|---------|
| 1   | 1:39.140 | 3.79   | 3.41   | excluded, off_track_surface, | 1  | 0    | 85.1   | 82.3   | 2.77 | 55.04  | 43.93 | 3.40   | 321  | 57     | 491     | unknown | 1/-     |
| 2   | 1:37.436 | 2.08   | 1.70   | valid representative         | 1  | 0    | 82.3   | 79.4   | 2.96 | 43.77  | 46.53 | 3.45   | 323  | 63     | 554     | unknown | 0/-     |
| 3   | 1:35.799 | 0.44   | 0.07   | valid representative         | 1  | 0    | 79.4   | 76.3   | 3.05 | 46.42  | 59.83 | 3.53   | 324  | 67     | 566     | unknown | 0/-     |
| 4   | 1:36.162 | 0.81   | 0.43   | valid representative         | 1  | 0    | 76.3   | 73.4   | 2.95 | 59.72  | 65.35 | 3.48   | 325  | 69     | 569     | unknown | 0/-     |
| 5   | 1:36.056 | 0.70   | 0.32   | valid representative         | 1  | 0    | 73.4   | 70.4   | 2.95 | 65.18  | 62.52 | 3.49   | 325  | 73     | 559     | unknown | 0/-     |
| 6   | 1:46.124 | 10.77  | 10.39  | valid representative         | 1  | 0    | 70.4   | 67.4   | 2.98 | 62.41  | 60.33 | 3.47   | 323  | 74     | 538     | unknown | 0/-     |
| 7   | 1:35.668 | 0.31   | -0.07  | valid representative         | 1  | 0    | 67.4   | 64.5   | 2.96 | 60.16  | 51.57 | 3.49   | 325  | 73     | 575     | unknown | 0/0.0   |
| 8   | 1:35.441 | 0.09   | -0.29  | valid representative         | 1  | 0    | 64.5   | 61.6   | 2.91 | 51.41  | 43.21 | 3.46   | 325  | 76     | 568     | unknown | 0/-     |
| 9   | 1:35.354 | 0.00   | -0.38  | valid representative         | 1  | 0    | 61.6   | 58.6   | 2.93 | 43.05  | 36.80 | 3.47   | 325  | 75     | 572     | unknown | 0/-     |
| 10  | 1:35.502 | 0.15   | -0.23  | valid representative         | 1  | 0    | 58.6   | 55.8   | 2.87 | 36.64  | 33.08 | 3.39   | 324  | 75     | 547     | unknown | 0/-     |
| 11  | 1:35.658 | 0.30   | -0.08  | valid representative         | 1  | 0    | 55.8   | 52.8   | 2.99 | 32.97  | 44.32 | 3.43   | 324  | 75     | 550     | unknown | 0/-     |
| 12  | 1:21.220 | -14.13 | -14.51 | excluded, lap_time_outside_s | 1  | 0    | 52.8   | 50.8   | 2.02 | 44.21  | 65.26 | 2.29   | 324  | 81     | 562     | unknown | 0/-     |
| 0   | 2:09.374 | 34.02  | 33.64  | excluded, lap_time_outside_s | -  | 0    | 88.0   | 85.1   | 2.90 | 100.00 | 55.25 | 3.05   | 310  | 46     | 426     | unknown | 0/1.0   |

# Appendix A - Channel Availability

Coverage reflects the indexed source manifest. Metrics use full-resolution source calculations where available; report traces may be downsampled.

| Channel               | Hz  | Rows   | Unit   | Mapped fields         | Usage          | Coverage |
|-----------------------|-----|--------|--------|-----------------------|----------------|----------|
| ABS                   | -   | 1      | -      | abs_active            | direct/derived | indexed  |
| ABSLevel              | -   | 1      | -      | abs_level             | direct/derived | indexed  |
| Ambient Temperature   | 1   | 1387   | C      | ambient_temp          | direct/derived | indexed  |
| AntiStall Activated   | -   | 1      | -      | anti_stall_active     | direct/derived | indexed  |
| Best LapTime          | -   | 7      | s      | best_lap_time         | direct/derived | indexed  |
| Best Sector1          | -   | 7      | s      | best_sector1          | direct/derived | indexed  |
| Best Sector2          | -   | 7      | s      | best_sector2          | direct/derived | indexed  |
| Brake Bias Rear       | -   | 1      | -      | brake_bias_rear       | direct/derived | indexed  |
| Brake Migration       | -   | 1      | -      | brake_migration       | direct/derived | indexed  |
| Brake Pos             | 50  | 69337  | %      | brake                 | direct/derived | indexed  |
| Brake Pos Unfiltered  | 50  | 69337  | %      | -                     | not used       | indexed  |
| Brake Thickness       | 10  | 13868  | %      | brake_thickness       | direct/derived | indexed  |
| Brakes Air Temp       | 50  | 69337  | C      | brake_air_temp        | direct/derived | indexed  |
| Brakes Force          | 50  | 69337  | %      | brake_force           | direct/derived | indexed  |
| Brakes Temp           | 50  | 69337  | C      | brake_temp            | direct/derived | indexed  |
| CloudDarkness         | -   | 1      | %      | cloud_darkness        | direct/derived | indexed  |
| Clutch Pos            | 50  | 69337  | %      | clutch                | direct/derived | indexed  |
| Clutch Pos Unfiltered | 50  | 69337  | %      | clutch_unfiltered     | direct/derived | indexed  |
| Clutch RPM            | 100 | 138673 | RPM    | clutch_rpm            | direct/derived | indexed  |
| Current LapTime       | -   | 12     | s      | current_lap_time      | direct/derived | indexed  |
| Current Sector        | -   | 39     | -      | sector                | direct/derived | indexed  |
| Current Sector1       | -   | 14     | s      | sector1               | direct/derived | indexed  |
| Current Sector2       | -   | 14     | s      | sector2               | direct/derived | indexed  |
| Engine Max RPM        | -   | 1      | RPM    | -                     | not used       | indexed  |
| Engine Oil Temp       | 7   | 9732   | C      | engine_oil_temp       | direct/derived | indexed  |
| Engine RPM            | 100 | 138673 | RPM    | rpm                   | direct/derived | indexed  |
| Engine Water Temp     | 7   | 9732   | C      | engine_water_temp     | direct/derived | indexed  |
| FFB Output            | 100 | 138673 | %      | ffb_output            | direct/derived | indexed  |
| Finish Status         | -   | 1      | -      | finish_status         | direct/derived | indexed  |
| Front3rdDeflection    | 100 | 138673 | m      | front_3rd_deflection  | direct/derived | indexed  |
| FrontFlapActivated    | -   | 1      | -      | front_flap_active     | direct/derived | indexed  |
| FrontRideHeight       | 100 | 138673 | m      | front_ride_height     | direct/derived | indexed  |
| Fuel Level            | 20  | 27735  | L      | fuel_liters           | direct/derived | indexed  |
| FuelMixtureMap        | -   | 6      | -      | fuel_mixture_map      | direct/derived | indexed  |
| G Force Lat           | 10  | 13868  | G      | g_force_lat           | direct/derived | indexed  |
| G Force Long          | 10  | 13868  | G      | g_force_long          | direct/derived | indexed  |
| G Force Vert          | 10  | 13868  | G      | g_force_vert          | direct/derived | indexed  |
| GPS Latitude          | 10  | 13868  | deg    | gps_latitude          | direct/derived | indexed  |
| GPS Longitude         | 10  | 13868  | deg    | gps_longitude         | direct/derived | indexed  |
| GPS Speed             | 10  | 13868  | m/s    | -                     | not used       | indexed  |
| GPS Time              | 100 | 138673 | s      | game_time             | direct/derived | indexed  |
| Gear                  | -   | 1006   | -      | gear                  | direct/derived | indexed  |
| Ground Speed          | 100 | 138673 | km/h   | speed_kph             | direct/derived | indexed  |
| Headlights State      | -   | 1      | On/Off | headlights            | direct/derived | indexed  |
| In Pits               | -   | 2      | -      | in_pits               | direct/derived | indexed  |
| Lap                   | -   | 13     | -      | lap_number            | direct/derived | indexed  |
| Lap Dist              | 10  | 13868  | m      | lap_distance          | direct/derived | indexed  |
| Lap Time              | -   | 13     | s      | last_lap_time         | direct/derived | indexed  |
| Last Sector1          | -   | 13     | s      | last_sector1          | direct/derived | indexed  |
| Last Sector2          | -   | 13     | s      | last_sector2          | direct/derived | indexed  |
| LastImpactMagnitude   | -   | 8      | -      | last_impact_magnitude | direct/derived | indexed  |

# Appendix A - Channel Availability

Coverage reflects the indexed source manifest. Metrics use full-resolution source calculations where available; report traces may be downsampled.

| Channel                 | Hz  | Rows   | Unit | Mapped fields               | Usage          | Coverage |
|-------------------------|-----|--------|------|-----------------------------|----------------|----------|
| LaunchControlActive     | -   | 1      | -    | -                           | not used       | indexed  |
| Minimum Path Wetness    | -   | 1      | %    | minimum_path_wetness        | direct/derived | indexed  |
| OffpathWetness          | -   | 1      | %    | offpath_wetness             | direct/derived | indexed  |
| OverheatingState        | 2   | 2774   | -    | overheating_state           | direct/derived | indexed  |
| Path Lateral            | 10  | 13868  | m    | path_lateral                | direct/derived | indexed  |
| Rear3rdDeflection       | 100 | 138673 | m    | rear_3rd_deflection         | direct/derived | indexed  |
| RearFlapActivated       | -   | 1      | -    | rear_flap_active            | direct/derived | indexed  |
| RearFlapLegalStatus     | -   | 1      | -    | rear_flap_legal             | direct/derived | indexed  |
| RearRideHeight          | 100 | 138673 | m    | rear_ride_height            | direct/derived | indexed  |
| Regen Rate              | 100 | 138673 | kW   | regen_rate                  | direct/derived | indexed  |
| RideHeights             | 100 | 138673 | m    | ride_height                 | direct/derived | indexed  |
| Sector1 Flag            | -   | 65     | -    | sector1_flag                | direct/derived | indexed  |
| Sector2 Flag            | -   | 58     | -    | sector2_flag                | direct/derived | indexed  |
| Sector3 Flag            | -   | 59     | -    | sector3_flag                | direct/derived | indexed  |
| SoC                     | 20  | 27735  | %    | soc                         | direct/derived | indexed  |
| Speed Limiter           | -   | 2      | -    | speed_limiter               | direct/derived | indexed  |
| Steering Pos            | 100 | 138673 | %    | steering                    | direct/derived | indexed  |
| Steering Pos Unfiltered | 100 | 138673 | %    | -                           | not used       | indexed  |
| Steering Shaft Torque   | 100 | 138673 | Nm   | steering_shaft_torque       | direct/derived | indexed  |
| SurfaceTypes            | -   | 2857   | -    | surface_type                | direct/derived | indexed  |
| Susp Pos                | 100 | 138673 | m    | suspension_position         | direct/derived | indexed  |
| TC                      | -   | 12359  | -    | tc_active                   | direct/derived | indexed  |
| TCCut                   | -   | 1      | -    | tc_cut                      | direct/derived | indexed  |
| TCLLevel                | -   | 1      | -    | tc_level                    | direct/derived | indexed  |
| TCSlipAngle             | -   | 1      | -    | tc_slip_angle               | direct/derived | indexed  |
| Throttle Pos            | 50  | 69337  | %    | throttle                    | direct/derived | indexed  |
| Throttle Pos Unfiltered | 50  | 69337  | %    | -                           | not used       | indexed  |
| Time Behind Next        | 2   | 2774   | s    | time_behind_next            | direct/derived | indexed  |
| Total Dist              | 10  | 13868  | m    | total_distance              | direct/derived | indexed  |
| Track Edge              | 10  | 13868  | m    | track_edge                  | direct/derived | indexed  |
| Track Temperature       | 1   | 1387   | C    | track_temp                  | direct/derived | indexed  |
| Turbo Boost Pressure    | 100 | 138673 | Pa   | turbo_boost_pressure        | direct/derived | indexed  |
| Tyres Wear              | 10  | 13868  | %    | tyre_wear                   | direct/derived | indexed  |
| TyresCarcassTemp        | 5   | 6934   | C    | tyre_temp_carcass           | direct/derived | indexed  |
| TyresCompound           | -   | 1      | -    | tyre_compound               | direct/derived | indexed  |
| TyresPressure           | 10  | 13868  | kPa  | tyre_pressure               | direct/derived | indexed  |
| TyresRimTemp            | 50  | 69337  | C    | tyre_temp_rim               | direct/derived | indexed  |
| TyresRubberTemp         | 10  | 13868  | C    | tyre_temp_rubber            | direct/derived | indexed  |
| TyresTempCentre         | 100 | 138673 | C    | tyre_temp, tyre_temp_centre | direct/derived | indexed  |
| TyresTempLeft           | 100 | 138673 | C    | tyre_temp_left              | direct/derived | indexed  |
| TyresTempRight          | 100 | 138673 | C    | tyre_temp_right             | direct/derived | indexed  |
| Virtual Energy          | 20  | 27735  | %    | virtual_energy              | direct/derived | indexed  |
| Wheel Speed             | 100 | 138673 | m/s  | wheel_speed                 | direct/derived | indexed  |
| WheelsDetached          | -   | 1      | -    | wheels_detached             | direct/derived | indexed  |
| Wind Heading            | 1   | 1387   | deg  | wind_heading                | direct/derived | indexed  |
| Wind Speed              | 1   | 1387   | m/s  | wind_speed                  | direct/derived | indexed  |
| Yellow Flag State       | -   | 1      | -    | yellow_flag_state           | direct/derived | indexed  |
| channelsList            | -   | 56     | -    | -                           | not used       | indexed  |
| eventsList              | -   | 42     | -    | -                           | not used       | indexed  |
| metadata                | -   | 11     | -    | -                           | not used       | indexed  |

# Appendix B-F - Events, Definitions & Limits

## Event log

Lap - - pit transit [information]: entry-to-exit 141.74 s (high confidence)

Lap 1 - off-track surface [warning]: 1 sampled contacts (high confidence)

Lap - - observed impact signal [information]: magnitude 1.0 (medium confidence)

## Derived metric definitions

- Representative pace: median of valid laps after pit, incomplete, flag, anomaly and shared quality exclusions.

- Fuel effect: Theil-Sen pairwise slope using at least six laps with fuel and pace; empirical, not a physical mass model.

- Tyre degradation: within-stint clean-pace slope where at least five comparable laps exist; correlated effects are disclosed.

- Theoretical best: sum of best valid sectors only when compatible sector splits exist; realistic potential remains separate.

- Traffic severity: derived from Time Behind Next when present; never treated as proof of an overtake.

## Confidence and limitations

High: directly measured or strongly supported. Medium: consistent multi-channel inference. Low: plausible with missing context. Unavailable: insufficient source evidence. Exact overtakes, setup changes, opponent strategy, tyre/brake target windows, aerodynamic coefficients, reaction time, damage and race-control state are not fabricated.

# Prioritized Actions

## PRIORITY 1 - Review pit execution

Evidence: 1 timed pit transits were detected. Action: Compare entry-to-exit phases and rehearse the slowest repeatable phase. Expected benefit: Potential benefit cannot be isolated without stationary timing. Confidence: low Validation: Record stationary phase and pit-cycle position state in the next race.

## PRIORITY 2 - Stabilize representative race pace

Evidence: Robust lap-time MAD is 0.308 s. Action: Use clean-air laps as the pace baseline and flag traffic/pit laps separately. Expected benefit: A clearer strategy baseline; numerical gain is not supportable from current data. Confidence: high Validation: Compare the next race stint at similar fuel and tyre age.